ModelIt!

Student Activity Pack

Designing a Mars Habitat That Won't Kill You

Modeling the Interdependent Systems of Extraterrestrial Human Survival

Grade 10L3 | NGSS HS-ESS1-4, HS-ETS1-1 | Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think are the most critical systems needed to keep humans alive on Mars?

2 How are the different life support systems in a Mars habitat connected to each other?

3 Draw a diagram showing what you think would happen if the power system on a Mars base failed during a dust storm.

4 Why is designing for Mars harder than designing for the International Space Station?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Radiation Shielding | Atmosphere Control | Water Recycling | Food Production | Energy Generation | Structural Integrity | Temperature Regulation | Waste Management | Psychological Well-Being

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Nominal Operations

All nine systems at optimal levels

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Dust Storm Crisis

Energy Generation drops by 50% for 30 days

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Water System Failure

Water Recycling drops from 98% to 60% efficiency

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Communication Relay — The system maintaining contact with Earth despite 4-24 minute signal delays, critical for receiving technical support and maintaining crew morale

Medical Capability — The onboard medical resources and crew training to handle injuries, illness, and surgical emergencies without the possibility of evacuation to Earth

Spare Parts Inventory — The stockpile of replacement components for critical systems, balanced against the enormous cost of launching mass from Earth to Mars

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What combination of engineering systems would actually keep humans alive on Mars — and what happens when one fails? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that Mars habitat design is fundamentally a systems engineering problem rather than a collection of independent challenges?**

2. **Which system failure in your model creates the fastest cascade to crew danger, and what does that tell you about design priorities?**

3. **Why did the model include Psychological Well-Being as a component — how does it interact with the engineering systems?**

4. **How would your model change if the habitat used nuclear power instead of solar as the primary energy source?**

5. **What are the limitations of modeling nine interconnected systems when the real habitat would have hundreds of subsystems?**

8 STEM Challenge: Design a Mars Habitat Crisis Response Protocol

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YOUR ENGINEERING MISSION

Design a crisis management protocol that prioritizes habitat systems during a 45-day global dust storm when energy generation drops to 40% of nominal capacity.

THE SCENARIO

Your Mars colony of eight crew members has just detected a global dust storm approaching that will reduce solar energy generation to 40% for approximately 45 days. The nuclear backup generator provides only 15% additional capacity. Your team must design a crisis protocol that determines which systems to reduce, which to maintain, and how to manage crew psychological health during the emergency.

Engineering Design Thinking

1. WHAT IS YOUR PRIORITY RANKING OF ALL NINE SYSTEMS AND WHAT EVIDENCE FROM YOUR MODEL SUPPORTS IT?
2. HOW MUCH CAN EACH NON-CRITICAL SYSTEM BE REDUCED BEFORE IT TRIGGERS CASCADING FAILURES?
3. WHAT CREW ROLES AND SCHEDULES WOULD YOU IMPLEMENT DURING EMERGENCY OPERATIONS?
4. HOW WOULD YOU MONITOR CREW PSYCHOLOGICAL STATE AND INTERVENE IF SOMEONE BEGINS BREAKING DOWN?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Aerospace Systems Engineers design and integrate the complex systems needed for space habitats and vehicles. They work at NASA, SpaceX, Lockheed Martin, and international space agencies, earning $90,000-$170,000/year. Human Factors Engineers who specialize in crew well-being in extreme environments earn $80,000-$140,000/year.